MOHAMMED SALIH

Backend-Focused Full-Stack Developer
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PROFESSIONAL SUMMARY

Backend-focused Full-Stack Developer with 4+ years of experience building production-ready web platforms and large-scale backend systems across full-time and independent work. Serve as CTO and Backend Lead at AfroDigital Innovation Labs, specializing in custom AI solutions. Built the backend for an integrated health service platform with a multi-vendor pharmacy marketplace, doctor consultations, and integrated digital payments, and for an AI-Director-Driven School Operating System that brings HR, finance, students, and parents into one platform with support for five languages. Own products end to end, mentoring and leading a team of 10 engineers delivering AI products for multiple clients. Founded BioMatch, an AI-powered organ donor matching platform recognized as a top startup at Lead Business 2025. Work mainly in TypeScript, Node.js (NestJS, Bun), and Python.

PROFESSIONAL EXPERIENCE AND LEADERSHIP

AfroDigital Innovation Labs PLC **Feb 2022 – Present**
CTO & Backend Lead *Full-time (Addis Ababa, Ethiopia)*

- Shipped custom ERPs, AI chatbots, workflow automations, and rich frontend UIs for 9+ client organizations and private-sector companies in education, healthcare, and marketing
- Led the development of products for 6 clients, choosing the tech stack, designing APIs and data models, and setting code review and deployment standards, while planning roadmaps closely with the CEO
- Wrote and peer-reviewed 70+ production-style coding challenges with reference solutions, which serve as training and evaluation data for custom AI models
- Reduced product shipping and delivery time by 50% (6 weeks to 3) by building a reusable NestJS backend foundation, now the company standard across client projects, and automating CI/CD (GitHub Actions, Docker)
- Lead developer for Tibeb, AfroDigital's own in-house integrated health platform in Addis Ababa: built a multi-vendor pharmacy marketplace where licensed pharmacies manage their own storefronts, inventory, and pricing, with prescription uploads, online doctor consultations, and automated vendor payouts
- Mentor and lead a team of 10 engineers developing 2 projects under a formal Agile lifecycle, using code reviews, architecture guidance, and pairing; onboard 5+ junior developers and guide their technical growth

BITS College School of Systems and Technology **May 2025 – Present**
Student Instructor & Backend Developer (Contract) *Part-time (Addis Ababa, Ethiopia)*

- Selected on the basis of academic performance (4.0 GPA) to teach junior software engineering students while still studying at the college
- Taught 40+ first-year students in the Python lab, guiding them from the concepts covered in lectures to writing working programs through hands-on coding practice
- Taught practical Data Structures & Algorithms to second-year students, guiding them to a deeper understanding of core concepts and stronger problem-solving skills
- Built backend features for a therapy management system that digitizes ABA therapy for children with autism, covering assessments, therapy plans, live session data collection, staff scheduling, and a parent portal (Ruby on Rails 8, PostgreSQL, Rodauth/JWT, RSpec)

PROJECTS

Parsim | *Inference & Context Reduction for Long-Horizon Agents*
Founder & Full-Stack Engineer · Personal product, in development *Jul 2026 – Present*

- A provider-agnostic inference and context-management layer that lets long-horizon AI agents run longer tasks at lower cost. Agents waste tokens reprocessing history that is no longer relevant; Parsim compresses that context while keeping task-relevant information, for faster responses and lower token usage.
- Cut agents' context tokens by 74–89% with garbage collection and up to 92% with background LLM summaries on pg-boss queues, across OpenAI, Anthropic, and Gemini, with prompt caching intact
- Built 45 API endpoints and 20 database tables with 860+ tests (568 unit, 237 end-to-end, 64 demo-agent)

YeneSchool | *AI-Director-Driven School Operating System* yeneschool.me
Founder & Lead Developer

- An AI-Director-Driven School Operating System for Ethiopian schools that brings HR, finance, academics, students, and parents into one platform, automating routine school work with AI in five languages.
- Led the team that built 900+ REST endpoints, 151 database tables, and 185 pages across 10 user roles on NestJS, PostgreSQL/Prisma, and Redis/BullMQ, with role-based access control

Agentum | *AI-Native Customer Support*

- An AI-native customer support platform that answers customers from a business's own documentation.
- Built document ingestion (PDF, CSV, text, and website scraping), knowledge sections with custom response tones, and an embeddable chat widget for third-party websites

BioMatch | *AI-Powered Organ Donor Matching*

- An AI-powered organ donor matching platform that automates months of manual compatibility checks.
- Founded and built the platform, matching donors with population genetics data and predictive analytics, tested end to end with Cypress and Selenium; recognized as a top startup at Lead Business 2025 (SNV Ethiopia)

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Ruby, PHP, Java, SQL, Bash

Backend & APIs: Node.js, NestJS, Bun, Express, Hono.js, Laravel, Ruby on Rails, FastAPI, REST, GraphQL, gRPC, tRPC, WebSockets, Microservices, OpenAPI/Swagger, Zod, OAuth, RBAC, Webhooks

Databases, Data & Messaging: PostgreSQL, MySQL, MongoDB, Redis, SQLite, Elasticsearch, Supabase, BigQuery, Snowflake, Prisma, Drizzle, Kafka, BullMQ

Frontend & Design Tools: React, Next.js, TanStack, Redux, Tailwind CSS, shadcn/ui, Figma, Framer, Webflow

Cloud, DevOps & Testing: AWS, Azure, Docker, Kubernetes, Terraform, GitHub Actions, CI/CD (Continuous Integration & Deployment), Linux, Nginx, Prometheus, Jest, Vitest, Supertest, Cypress, Selenium, Unit Testing, Automated Testing

AI, Automation & Integrations: OpenAI API, Claude API, LangChain, RAG, MCP, AI Agents, LLM Training Data (SFT/RL), n8n, Zapier, Make, GoHighLevel, Telegram Bot API & Mini Apps, Google Workspace APIs

Engineering Practices: System Design, API Design, Database Design, Performance Optimization, Test-Driven Development, Code Review, Technical Mentoring, Agile/Scrum, Git

EDUCATION AND CERTIFICATIONS

BITS College School of Systems and Technology | Bachelor of Software Engineering, GPA 4.0 / 4.0 Oct 2024 – Present
Entrepreneurship Training Workshop (ETW) | Empretec and Entrepreneurship Development Institute (EDI) Sep 2026
Design Your Venture (DYV) | Entrepreneurship Development Institute Sep 2026
Certificate of Recognition — Top Startup (BioMatch) | Lead Business 2025, SNV Ethiopia Nov 2025
Full Stack Web Development Certification | Evangadi Tech Dec 2024

LANGUAGES

English (Fluent), Amharic (Native)